Weak Maximum Principle and Dirichlet Uniqueness

An End-to-End Meta-Math Walkthrough

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Abstract

A non-strict differential inequality does not directly rule out an interior maximum. Add a small strictly subharmonic quadratic, then remove the perturbation. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PDE-01 / Partial Differential Equations / Report v1

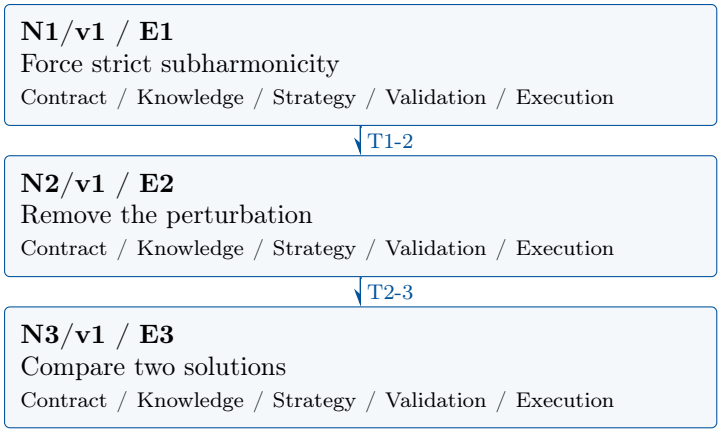
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $\Omega \subset \mathbb{R}^d$ be a nonempty bounded open set and $u \in C^2(\Omega) \cap C(\overline{\Omega})$. If $\Delta u \geq 0$ in Ω, prove $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$. Deduce uniqueness in this class for a prescribed Laplacian and prescribed continuous boundary values. No boundary smoothness or connectedness is needed for this weak principle.*

A. Foundations and permitted imports

Use compact attainment of extrema on closed bounded subsets of Euclidean space. At an interior local maximum of a C^2 function, each pure second derivative is nonpositive, hence so is the Laplacian. The bounded open set has compact nonempty boundary. This is a maximum-location result, not the strong maximum principle about interior constancy.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $\max_{\overline{\Omega}} u_\varepsilon = \max_{\partial\Omega} u_\varepsilon$.

Receiving activity. N2.1

Compatibility. E1/v1 and $R^2 = \max_{\overline{\Omega}} \|x\|^2 < \infty$. Check: The error bound must be independent of the point x. Boundedness of the domain provides that control.

T2-3: N2/E2/v1 to N3. $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$.

Receiving activity. N3.1

Compatibility. E2/v1 and u, v with equal Laplacians and equal boundary values in the stated class. Check: Both signs are needed: one application gives only one ordering.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Force strict subharmonicity

Object and role. Use the positive Laplacian of the quadratic to contradict the second-derivative test at an interior maximum. **Local obligation:** $\max_{\overline{\Omega}} u_\varepsilon = \max_{\partial\Omega} u_\varepsilon$.

Inputs. $u_\varepsilon(x) = u(x) + \varepsilon\|x\|^2$ with $\varepsilon > 0$.

C — Contract

Publish E1/v1 only after establishing this obligation: $\max_{\overline{\Omega}} u_\varepsilon = \max_{\partial\Omega} u_\varepsilon$.

K — Knowledge

$u_\varepsilon(x) = u(x) + \varepsilon\|x\|^2$ with $\varepsilon > 0$. Retain the stated assumptions and import versions.

S — Strategy

Use the positive Laplacian of the quadratic to contradict the second-derivative test at an interior maximum.

V — Validation

The maximum exists on the compact closure; $\Delta u_\varepsilon \geq 2d\varepsilon > 0$ rules out every interior maximizing point.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 The perturbed function is continuous on the compact closure, so it attains a maximum. At an interior maximum its Hessian has nonpositive diagonal entries, giving $\Delta u_\varepsilon \leq 0$.

N1.2 But $\Delta u_\varepsilon = \Delta u + 2d\varepsilon > 0$ everywhere inside. Hence no maximizing point is interior; a maximizing point lies on the boundary, proving the export.

Export E1/v1. $\max_{\overline{\Omega}} u_\varepsilon = \max_{\partial\Omega} u_\varepsilon$.

Acceptance gate. The maximum exists on the compact closure; $\Delta u_\varepsilon \geq 2d\varepsilon > 0$ rules out every interior maximizing point.

Targeted repair. If $\Delta u \geq 0$ alone is claimed to contradict an interior maximum, restore the strictly positive perturbation first. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Remove the perturbation

Object and role. Bound the perturbation uniformly so its effect vanishes as $\varepsilon \downarrow 0$. **Local obligation:** $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$.

Inputs. E1/v1 and $R^2 = \max_{\overline{\Omega}} \|x\|^2 < \infty$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$.

K — Knowledge

E1/v1 and $R^2 = \max_{\overline{\Omega}} \|x\|^2 < \infty$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Bound the perturbation uniformly so its effect vanishes as $\varepsilon \downarrow 0$.

V — Validation

The error bound must be independent of the point x. Boundedness of the domain provides that control.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For every $x \in \overline{\Omega}$, E1 gives $u(x) \le u_\varepsilon(x) \le \max_{\partial\Omega} u + \varepsilon R^2$. Let $\varepsilon \downarrow 0$ to obtain $u(x) \le \max_{\partial\Omega} u$.

N2.2 Taking the maximum over the closure gives one inequality, and inclusion of the boundary in the closure gives the reverse. This proves the maximum principle.

Export E2/v1. $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$.

Acceptance gate. The error bound must be independent of the point x. Boundedness of the domain provides that control.

Targeted repair. If the quadratic perturbation is removed without a uniform bound, record R before taking the limit. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Compare two solutions

Object and role. Apply the maximum principle to the difference and its negative. **Local obligation:** $u = v$ on $\overline{\Omega}$, so the classical Dirichlet problem has at most one solution in this class.

Inputs. E2/v1 and u, v with equal Laplacians and equal boundary values in the stated class. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: $u = v$ on $\overline{\Omega}$, so the classical Dirichlet problem has at most one solution in this class.

K — Knowledge

E2/v1 and u, v with equal Laplacians and equal boundary values in the stated class. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Apply the maximum principle to the difference and its negative.

V — Validation

Both signs are needed: one application gives only one ordering.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 The difference $w = u - v$ has $\Delta w = 0$ and zero boundary values. Apply E2 to w to get $w \leq 0$.

N3.2 Apply it to $-w$, whose Laplacian also vanishes, to get $-w \leq 0$. Thus $w = 0$. This proves uniqueness without asserting existence for arbitrary data.

Export E3/v1. $u = v$ on $\overline{\Omega}$, so the classical Dirichlet problem has at most one solution in this class.

Acceptance gate. Both signs are needed: one application gives only one ordering.

Targeted repair. If uniqueness is inferred from only $w \leq 0$, apply the same argument to $-w$ before exporting equality. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Force strict subharmonicity

1. **N1.1.** The perturbed function is continuous on the compact closure, so it attains a maximum. At an interior maximum its Hessian has nonpositive diagonal entries, giving $\Delta u_\varepsilon \leq 0$.

2. **N1.2.** But $\Delta u_\varepsilon = \Delta u + 2d\varepsilon > 0$ everywhere inside. Hence no maximizing point is interior; a maximizing point lies on the boundary, proving the export.

B. N2 — Remove the perturbation

1. **N2.1.** For every $x \in \overline{\Omega}$, E1 gives $u(x) \leq u_\varepsilon(x) \leq \max_{\partial\Omega} u + \varepsilon R^2$. Let $\varepsilon \downarrow 0$ to obtain $u(x) \leq \max_{\partial\Omega} u$.

2. **N2.2.** Taking the maximum over the closure gives one inequality, and inclusion of the boundary in the closure gives the reverse. This proves the maximum principle.

C. N3 — Compare two solutions

1. **N3.1.** The difference $w = u - v$ has $\Delta w = 0$ and zero boundary values. Apply E2 to w to get $w \leq 0$.

2. **N3.2.** Apply it to $-w$, whose Laplacian also vanishes, to get $-w \leq 0$. Thus $w = 0$. This proves uniqueness without asserting existence for arbitrary data.

□

V. WORKED ORCHESTRATION

On the unit disk in $\mathbb{R}^2$, $u(x, y) = x^2 + y^2$ has $\Delta u = 4$ and maximum one on the boundary. For harmonic boundary data given by the restriction of x, the function $u(x, y) = x$ is a solution, and the comparison argument proves it is the only one in the stated class.

A. Boundary and hypothesis audit

Boundedness cannot simply be dropped. On the upper half-plane, $u(x, y) = y$ is harmonic and continuous up to the boundary, where it is zero, but it is positive and unbounded inside. Compact maximum attainment and the uniform perturbation bound both fail.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 argument: an interior maximum contradicts $\Delta u \geq 0$. A constant function has interior maxima and $\Delta u = 0$, so the claimed contradiction is invalid. **Repair:** use u_ε to obtain strict positivity, prove boundary attainment there, and then remove the perturbation. This fixes the proof while leaving the target weak maximum principle unchanged; recheck E2 and the uniqueness corollary.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Develop comparison for nonzero forcing or the strong maximum principle with its additional connectedness and argument.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Leon Simon, Lectures on PDE, for elliptic, weak, and parabolic methods. Classical calculus and the explicitly stated Hilbert-space imports are used below; the displayed PDE arguments are developed in full.